NYC GREEN TAXI DEMAND AND OPERATIONS ANALYSIS

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Outlines

01 Introduction

Introduces the project by outlining the background context, identifying key business problems, and defining the objectives to be achieved through the analysis.

02 Data Preparation

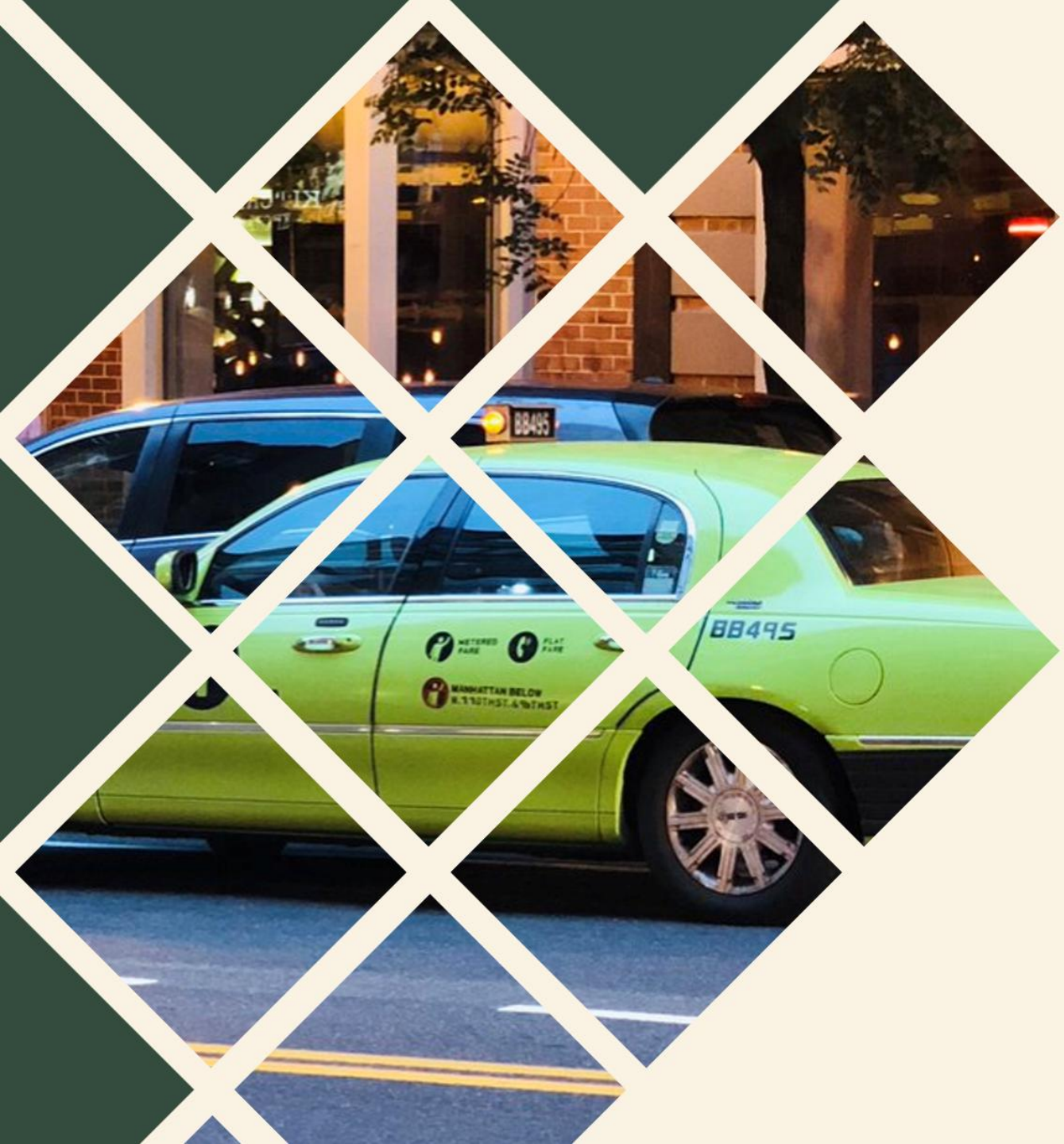
Details the processes of understanding the dataset's structure and contents, followed by cleaning steps to ensure the data is accurate, consistent, and ready for analysis.

03 Data Analysis

Focuses on analyzing key aspects of the dataset, including trip demand trends, spatial distribution, fare-related factors, and passenger behavior to uncover insights that drive operational strategies.

04 Conclusion and Recommendations

The conclusion highlights key findings, and the recommendations offer strategies to optimize operations and improve customer experience.



INTRODUCTION

Introduces the project by outlining the background context, identifying key business problems, and defining the objectives to be achieved through the analysis.

Background

NYC Transportation & TLC

- **Transportation** plays a **crucial role** in the daily lives of **over 8 million** New York City (**NYC**) residents and visitors who rely on an efficient and accessible system.
- The Taxi and Limousine Commission (**TLC**), established in **1971**, regulates all **for-hire transportation services**.
- **TLC** oversees **yellow taxis**, **green taxis**, for-hire vehicles **(FHVs**), **commuter vans**, and **paratransit**.

Purpose of Green Taxis

- Introduced in **2013** to expand coverage in underserved **outer boroughs**.
- Green Taxis operate in **Brooklyn, Queens, the Bronx, Staten Island**, and **upper Manhattan.**
- **Restricted** from picking up street hails in **central Manhattan** (below 110th and 96th Streets) unless pre-arranged.

Why Analysis is Needed

- TLC **collects trip data** from taxi meters, dispatch systems, and app-based ride services.
- **Raw data** alone is **insufficient** without actionable insights.
- **Analysis** helps optimizing **service availability** with **demand**, promote **fair** and **consistent pricing**, and support **more efficient fleet deployment** across the city.

Project Focus & Benefits

- Focuses on **Green Taxi trip records** to gain deeper operational insights.
- Analyze **peak demand times, key pickup locations, fare-related factors, and passenger behavior.**
- Provides data-driven recommendations for **service optimization** and an **improved passenger experience**.

Business Problems and Objectives

BUSINESS PROBLEMS

Green Taxis were introduced to improve transportation access in NYC's outer boroughs and upper Manhattan. However, there are still challenges in optimizing their operations. This project aims to address the following **key questions**:

01 **When is the highest demand for Green Taxis?**
Understanding when people use Green Taxis the most can help improve vehicle availability, reduce wait times, and enhance overall service efficiency.

02 **Where is trip demand concentrated geographically?**
Identifying high-demand pickup areas helps ensure that drivers are positioned in the right places to meet passenger needs, reduce idle time, and maximize ride opportunities.

03 **What factors influence taxi fares?**
Exploring how trip features like distance and duration relate to fare amounts can offer insights into pricing fairness and help optimize earning potential for drivers.

04 **How do passengers typically travel and pay?**
Knowing how many passengers usually share a ride and which payment methods are preferred can guide service improvements and operational planning.

OBJECTIVES

- Determine peak demand periods
- Identify high-demand locations
- Analyze fare-related factors
- Understand passenger behavior

DATA PREPARATION

Details the processes of understanding the dataset's structure and contents, followed by cleaning steps to ensure the data is accurate, consistent, and ready for analysis.

Data Understanding

Feature	Description
VendorID	A code indicating the LPEP provider. 1 = Creative Mobile Technologies, LLC; 2 = VeriFone Inc.
lpep_pickup_datetime	The date and time when the meter was engaged (pickup time).
lpep_dropoff_datetime	The date and time when the meter was disengaged (drop-off time).
store_and_fwd_flag	Y = store and forward trip; N = not stored before sending.
RateCodeID	Final rate code: 1 = Standard, 2 = JFK, 3 = Newark, 4 = Nassau/Westchester, 5 = Negotiated, 6 = Group
PULocationID	Pickup location ID (TLC Taxi Zone).
DOLocationID	Drop-off location ID (TLC Taxi Zone).
passenger_count	Number of passengers (driver-entered).
trip_distance	Distance in miles (from taximeter).
fare_amount	Fare calculated by the meter (starting at $3.00 base).
extra	Night ($1) or rush-hour ($2.50) surcharge.
mta_tax	$0.50 MTA tax based on metered rate.
tip_amount	Credit card tips only (cash tips excluded).
tolls_amount	Total tolls paid.
ehail_fee	Fee for e-hailing.
improvement_surcharge	$1.00 surcharge per hailed trip.
total_amount	Total amount charged (excluding cash tips).
payment_type	1 = Credit, 2 = Cash, 3 = No charge, 4 = Dispute, 5 = Unknown, 6 = Voided trip.
trip_type	1 = Street-hail, 2 = Dispatch.
congestion_surcharge	Fee for congestion zones ($2.75 Green Taxi, $2.50 Yellow, $0.75 shared rides).

Additional Dataset

Feature	Description
LocationID	A distinct numerical code representing each taxi zone.
Borough	The borough in which the taxi zone is situated, such as Manhattan, Queens, Brooklyn, Bronx or Staten Island.
Zone	The particular neighborhood or area within the borough.
service_zone	The classification of the service zone (e.g., 'Boro Zone' for Green Taxis, 'Yellow Zone' for Yellow Taxis, 'EWR' for Newark Airport).

Scope and Limitations

- This analysis uses **January 2023** trip data from NYC Green Taxis (LPEP).
- It **does not consider** seasonal trends **beyond** the dataset **timeframe**, external factors (e.g., weather, traffic), or compare with ride-hailing apps like Uber or Lyft.

Data Cleaning

1

Missing Values

- **Dropped the column**: ehail_fee (100% missing).
- **Filled rows with mode**: trip_type, store_and_fwd_flag, RatecodeID, payment_type.
- **Filled rows with median**: passenger_count, congestion_surcharge.
- **Dropped rows with missing values**: Borough_pickup, Zone_pickup, service_zone_pickup, Borough_dropoff, Zone_dropoff, service_zone_dropoff.

***Rows**: 68,211 → 67,499*
***Columns**: 26 → 25*
No missing values remain.

2

Incorrect Data Types

- **Converted to datetime**: lpep_pickup_datetime, lpep_dropoff_datetime.
- **Converted to integer**: passenger_count.
- **Converted to categorical**: VendorID, RatecodeID, PULocationID, DOLocationID, store_and_fwd_flag, payment_type, trip_type.
- **Converted to boolean**: store_and_fwd_flag ('Y' → True, 'N' → False).

All incorrect data types have been converted appropriately.

3

Duplicate Data

Found and dropped 168 duplicate rows based on **critical columns**: VendorID, RatecodeID, payment_type, trip_type, lpep_pickup_datetime, lpep_dropoff_datetime, PULocationID, DOLocationID, passenger_count, trip_distance.

***Rows**: 67,499 → 67,331*
Only unique trip records retained.

Data Cleaning

4

Inconsistent Data

- **Dropped rows with negative values**:
  fare_amount, total_amount.
- **Replaced negative values with 0**:
  extra, tip_amount, congestion_surcharge.
- **Corrected values**:
  mta_tax = 0.50, improvement_surcharge = 1.00.
- **Replaced 0 with mode/median**:
  passenger_count → mode, trip_distance → median.
- **Replaced invalid values**:
  RatecodeID (values outside 1–6).
- **Dropped rows** with:
  >6 passenger_count, **invalid pickup/dropoff datetime**, **lpep_pickup_datetime = lpep_dropoff_datetime**, **financial values outside valid range** (e.g., fare_amount < $3, tip_amount > $100).
- **Fixed typos and formatting in categorical columns** (e.g., lowercased values).

***Rows**: 67,331 → 65,094*
All inconsistencies resolved.

5

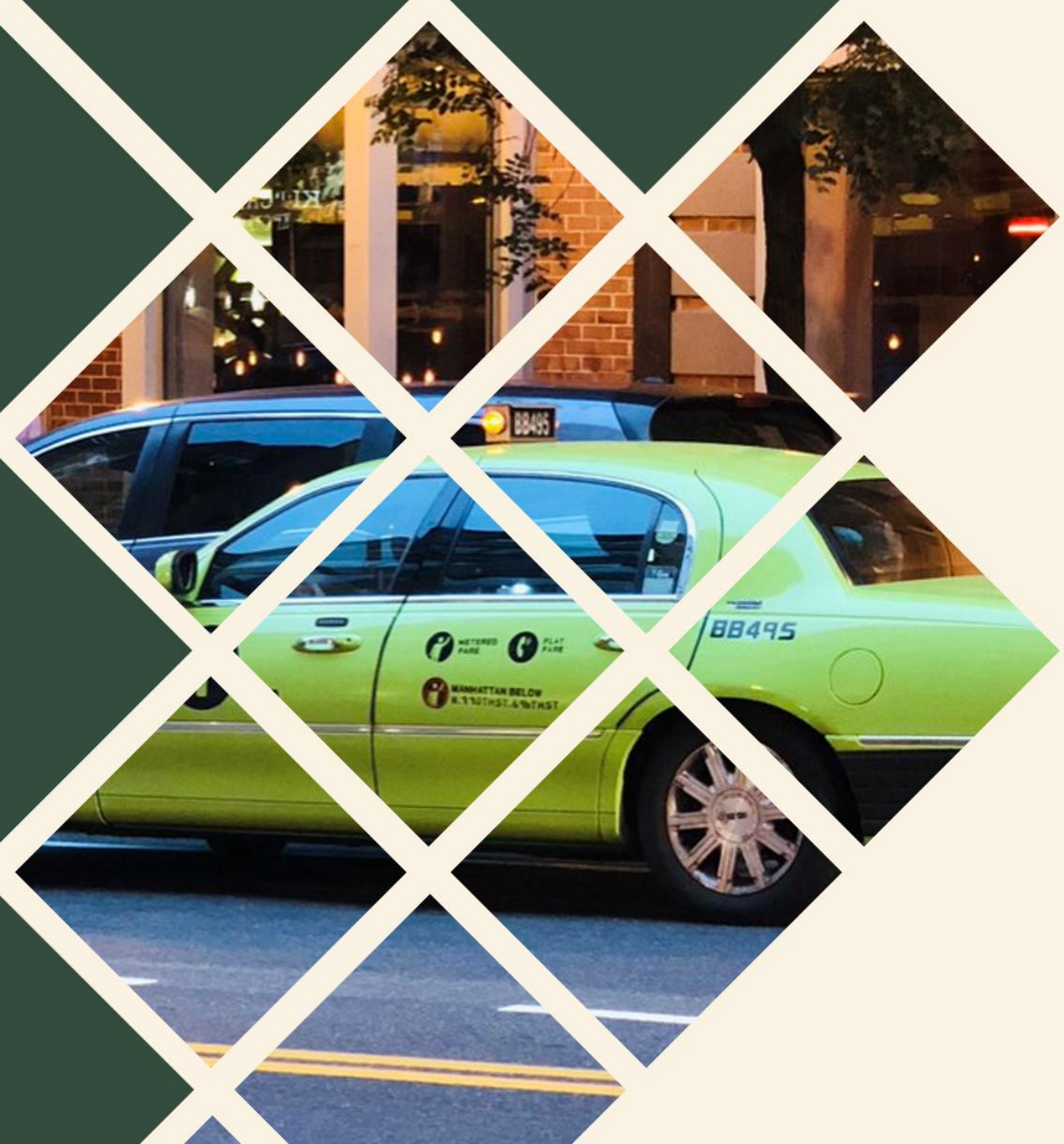
Outliers

- **Detected outliers**:
  passenger_count, trip_distance, fare_amount, extra, tip_amount, tolls_amount, total_amount, congestion_surcharge.
- **Dropped severe outliers**:
  trip_distance, fare_amount, tip_amount, tolls_amount, total_amount.
- **Retained acceptable values**:
  passenger_count, extra, congestion_surcharge.

***Rows**: 65,094 → 56,493*
Improved data distribution.

CLEANED DATASET OVERVIEW

Final Rows: 56,493
Final Columns: 25
Dataset is clean, structured, and ready for analysis.

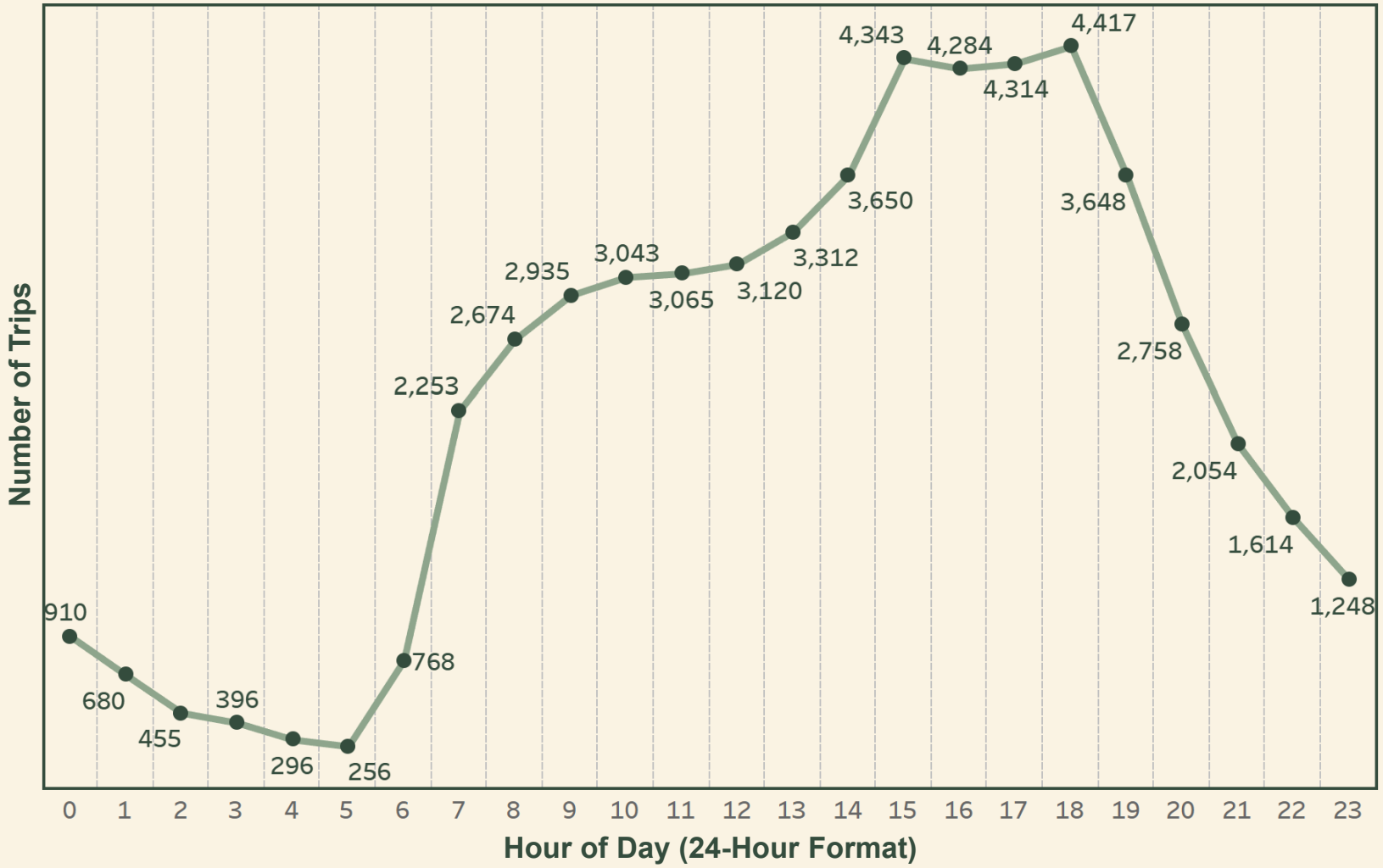


DATA ANALYSIS

Focuses on analyzing key aspects of the dataset, including trip demand trends, spatial distribution, fare-related factors, and passenger behavior to uncover insights that drive operational strategies.

Trip Demand Overtime Analysis

TRIP DEMAND BY HOUR OF DAY



Insights

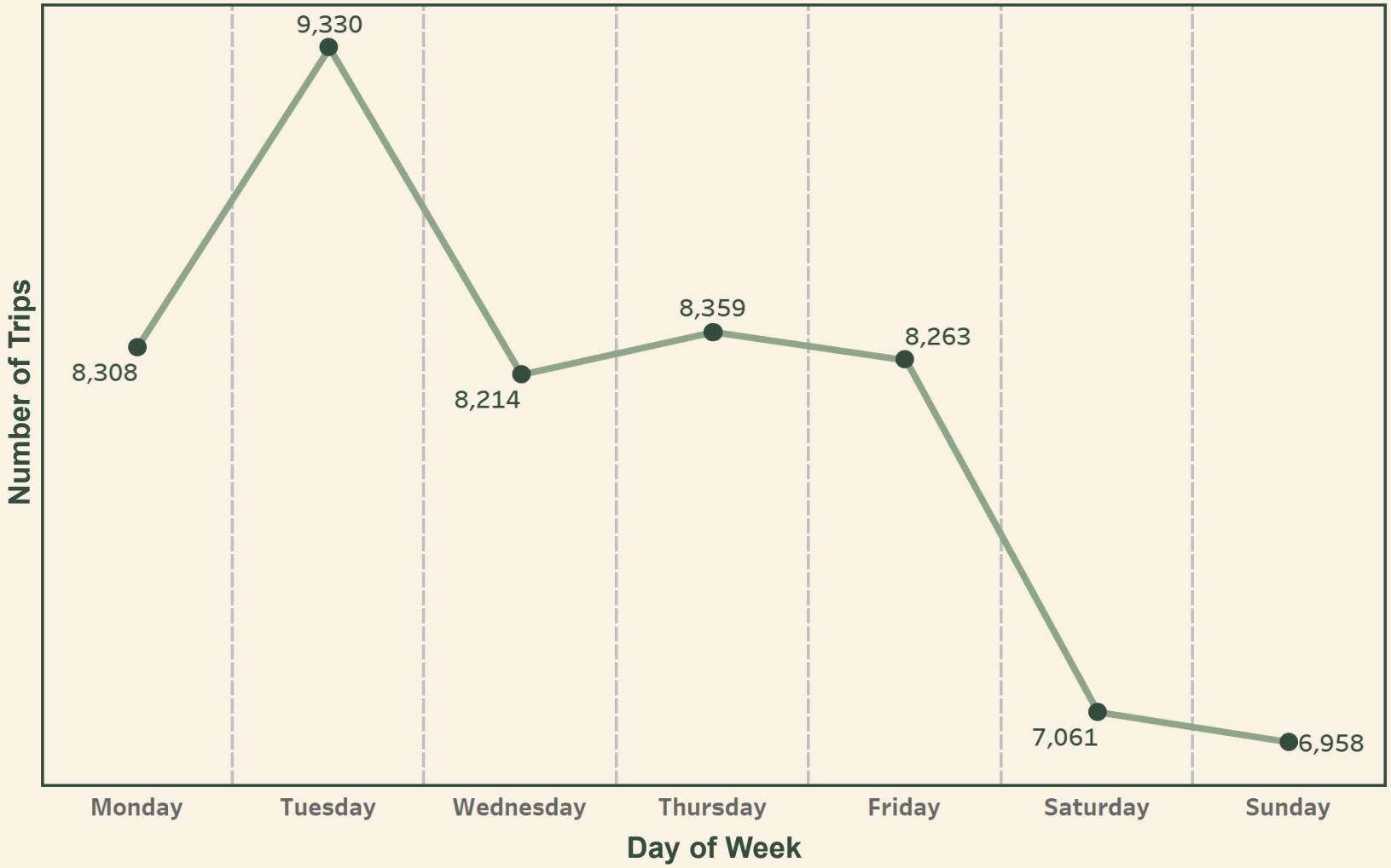
- **Evening Peak (3–6 PM):** Busiest period, peaking at **4,417 trips** at **6 PM** from work, school, and leisure transitions. Key for full fleet deployment and optimized break scheduling.
- **Morning Commute (6–10 AM):** Fast rise in demand (from **768** to **3,000+** trips/hour), driven by structured travel. Needs early vehicle readiness.
- **Midday (11 AM–2 PM):** Stable demand (**3,065–3,650**/hour), ideal for consistent service.
- **Evening Decline (7–11 PM):** Demand falls to **1,248** trips/hour, allowing fleet reduction and reallocation.
- **Overnight (12–5 AM):** Lowest demand, best for rest, maintenance, and prep for AM peak.

Trip Demand Overtime Analysis

Insights

- **Tuesday Peak:** Highest weekday demand **(9,330 trips)**, driven by in-person meetings and office attendance, making it the strongest operational day.
- **Weekday Stability:** Consistent high demand **(8,214–9,330 trips)** driven by commuting and school travel, ideal for predictable shift planning.
- **Weekend Drop:** Significant demand **decrease**, especially on **Sunday** (25% lower than Tuesday), reflecting leisure travel; consider scaling down services.
- **Saturday Outpaces Sunday:** Slightly higher demand on Saturday, offering opportunities in nightlife and leisure areas.
- **Weekday Consistency:** Monday, Wednesday, Thursday, and Friday show stable demand, making weekdays reliable for operational planning.

TRIP DEMAND BY DAY OF WEEK



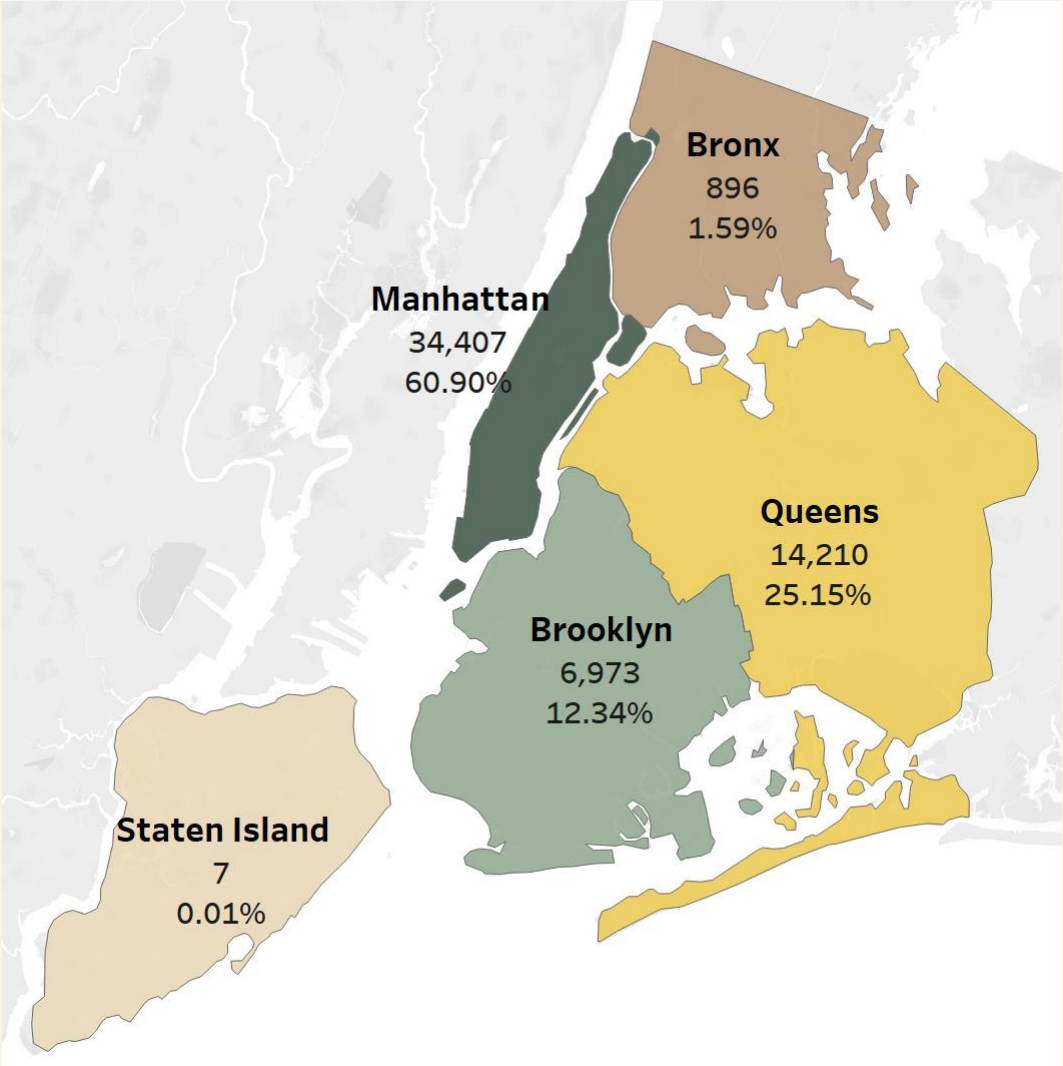
Trip Demand Overtime Analysis

HEATMAP: HOURLY TRIP DEMAND BY DAY OF WEEK

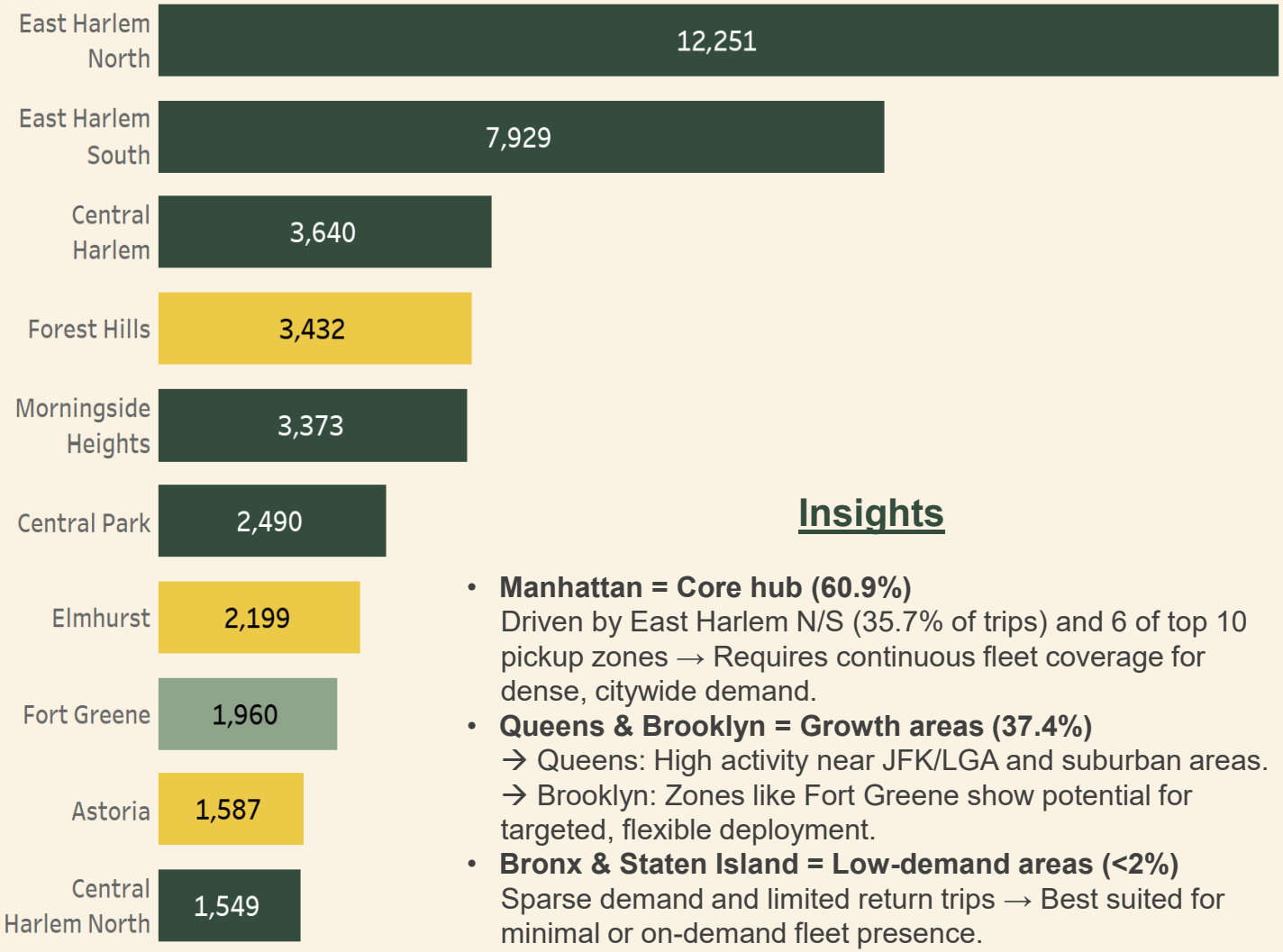
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Monday	100	59	39	61	39	36	110	356	394	462	451	475	493	546	544	670	649	658	665	513	350	293	206	139
Tuesday	77	49	34	29	12	41	171	502	609	587	546	499	524	504	601	685	726	710	708	599	436	307	227	147
Wednesday	82	55	25	19	14	31	141	406	481	489	480	424	401	450	513	634	631	669	679	578	397	267	207	141
Thursday	87	60	43	26	30	32	137	399	483	512	468	456	432	441	534	603	599	684	715	522	417	291	227	161
Friday	128	76	39	24	15	30	120	406	432	441	444	414	400	435	558	649	617	629	652	539	398	312	275	230
Saturday	197	156	99	81	70	35	37	95	162	229	344	397	415	481	435	577	543	474	515	460	402	303	277	277
Sunday	239	225	176	156	116	51	52	89	113	215	310	400	455	455	465	525	519	490	483	437	358	281	195	153

Trip Demand by Location Analysis

TRIP DEMAND BY PICKUP BOROUGH



TOP ZONES BY TRIP DEMAND



Insights

- **Manhattan = Core hub (60.9%)**
  Driven by East Harlem N/S (35.7% of trips) and 6 of top 10 pickup zones → Requires continuous fleet coverage for dense, citywide demand.
- **Queens & Brooklyn = Growth areas (37.4%)**
  → Queens: High activity near JFK/LGA and suburban areas.
  → Brooklyn: Zones like Fort Greene show potential for targeted, flexible deployment.
- **Bronx & Staten Island = Low-demand areas (<2%)**
  Sparse demand and limited return trips → Best suited for minimal or on-demand fleet presence.

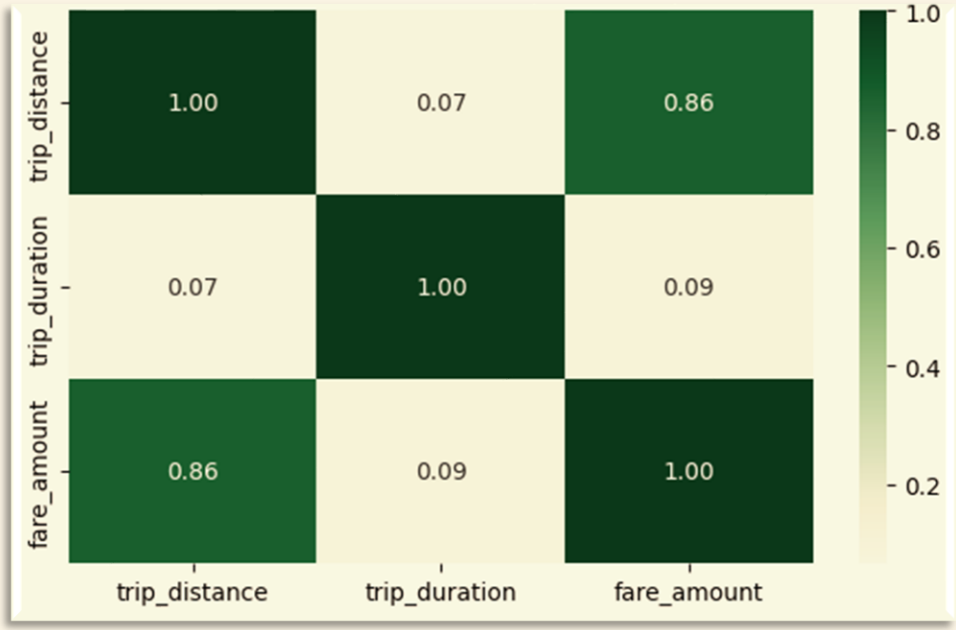
Trip Demand by Location Analysis

HEATMAP: HOURLY TRIP DEMAND IN TOP ZONES

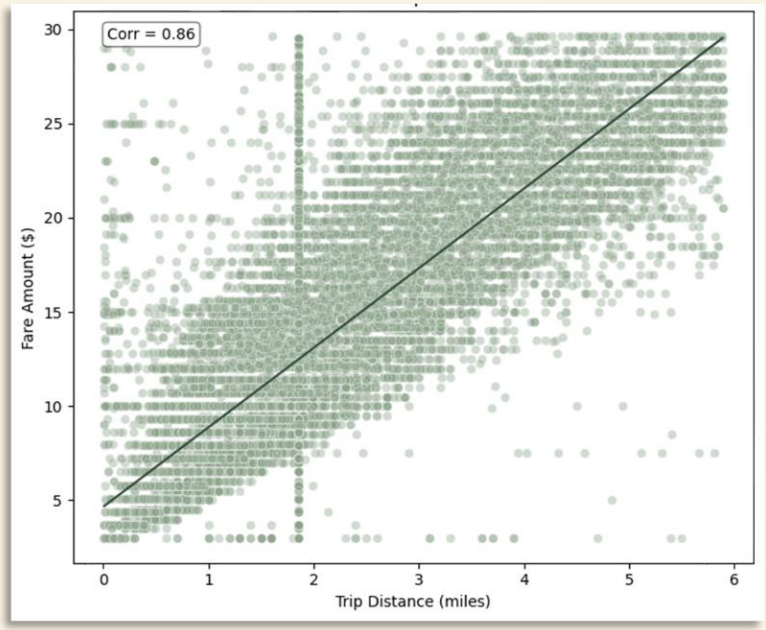
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
East Harlem North	68	52	16	10	6	64	477	1,111	1,027	973	751	743	647	662	704	788	797	712	810	598	494	341	246	154
East Harlem South	78	53	28	8	8	26	64	254	289	375	447	395	439	457	575	733	713	790	691	557	337	263	203	146
Central Harlem	43	29	15	10	7	5	26	192	256	227	252	254	244	282	290	350	239	191	195	158	122	100	76	77
Forest Hills	116	63	34	25	11	11	21	54	84	103	117	135	156	194	229	302	281	281	254	248	226	187	179	121
Morningside Heights	10	7	3	4	1	6	11	88	175	209	231	234	231	238	270	339	307	244	243	189	135	100	59	39
Central Park	2		2			2	10	48	85	158	144	154	174	153	170	171	282	267	215	210	122	69	31	21
Elmhurst	56	47	30	43	42	12	7	15	16	27	37	65	75	104	103	153	184	236	281	257	170	100	78	61
Fort Greene	14	15	5	1				2	8	24	67	130	157	147	187	202	178	172	218	206	112	73	32	10
Astoria	39	43	14	16	10	13	21	25	38	46	70	96	89	112	101	99	132	120	115	130	84	54	67	53
Central Harlem North	30	16	16	12	7	2	18	71	94	96	104	105	100	129	139	138	97	73	83	53	48	34	47	37

Fare-Related Factors Analysis

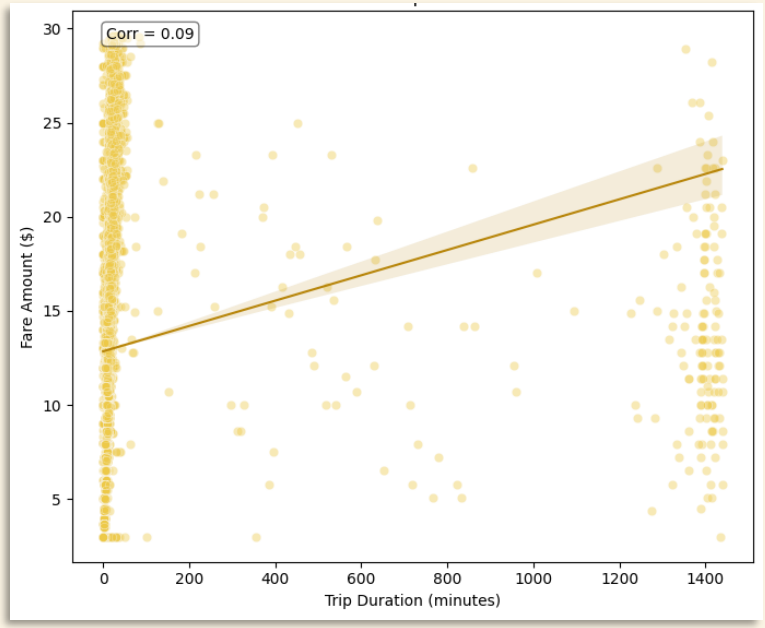
Correlation Matrix: Trip Distance, Duration, Fare



Fare vs Trip Distance



Fare vs Trip Duration



Insights

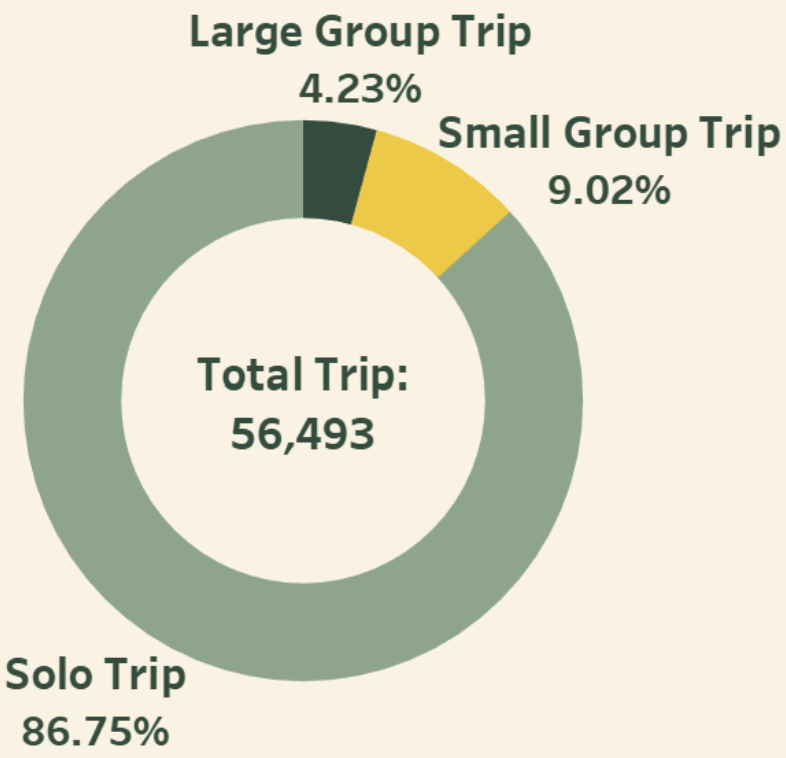
- **Trip Distance Strongly Predicts Fare (Corr = 0.86)**: Fares rise consistently with distance.
- **Trip Duration Has Weak Impact on Fare (Corr = 0.09)**: Longer trips ≠ higher fares. Time-based fees apply only in traffic/idle, not during normal trips.
- **Distance & Duration Are Minimally Correlated (Corr = 0.07)**: NYC traffic causes short trips to take long, disrupting predictability.

TLC fare structure:

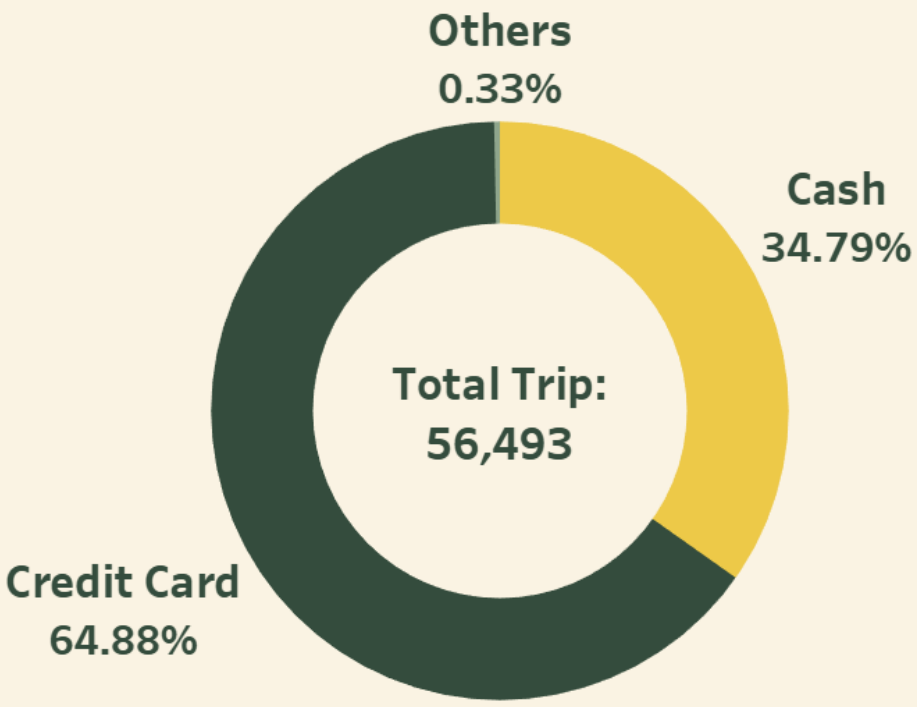
- **$3.00** base fare
- **$0.70 per 1/5 mile** above 12 mph
- **$0.70 per 60 sec** only when idle or <12 mph

Passenger Behaviour Analysis

TOTAL TRIPS BY PASSENGER GROUP



TOTAL TRIPS BY PAYMENT METHODS



CONCLUSION & RECOMMENDATIONS

The conclusion highlights key findings, and the recommendations offer strategies to optimize operations and improve customer experience.

Conclusion

TIME-BASED DEMAND PATTERNS

1. **Tuesdays show the highest trip volume**, peaking at 8 AM, 4 PM, and 5 PM → Top priority day for fleet deployment.
2. **Weekday peaks**: Consistent **7–9 AM** and **3–6 PM** surges from commuters → Maximize coverage with focused shift planning and incentives.
3. **12–5 AM and Sunday = lowest activity,** especially at 5 AM → Ideal for maintenance & driver rest.

FARE STRUCTURE INSIGHTS

1. **Distance is the strongest fare predictor (corr = 0.86)** → This aligns with NYC's distance-based pricing model and makes it ideal for forecasting and incentive design.
2. **Duration has minimal impact on fare (corr = 0.09)** → It only plays a role during idle time or low-speed conditions, making it unreliable for modeling or operational planning.
3. **Distance has a weak correlation with duration (corr = 0.07)** → This highlights the unpredictability of NYC traffic and reinforces the need for a distance-first strategy.

SPATIAL DEMAND CONCENTRATION

1. **Manhattan is the core hub with over 60% pickup demand,** led by **East Harlem North & South (1/3 of all trips)** → Needs continuous coverage and peak-time dispatch (7 AM & 5 PM).
2. **Bronx & Staten Island = low-demand (<2%)** → Best served with on-demand or minimal fleet due to inconsistent volume.

PASSENGER BEHAVIOUR PATTERNS

1. **Solo rides dominate (86.75%),** favoring a small, flexible fleet; ride-sharing can be boosted via apps, incentives & green messaging.
2. **Few multi-passenger trips (9.02% for small group; 4.23% for large group)** → larger vehicles underused, best for airports/events.
3. **Card payments lead (64.88%)** → enables loyalty, insights & security.
4. **Cash still common (34.79%), tied to tourists & older riders** → creates data gaps; recommend gradual digital shift with incentives & support.

Recommendations

TIME-BASED DEMAND OPTIMIZATION

1. **Maximize fleet** during **weekday peaks** (6–10 AM & 3–6 PM), especially **Tuesdays**.
2. **Reallocate resources to match trends** (e.g., Friday evenings, weekend 10 AM–6 PM for leisure areas).
3. **Scale down during low-demand hours** (12–5 AM, Sundays) to cut costs. **Use Sundays for maintenance**, training, and updates.

FARE MODELING AND REVENUE STRATEGY

1. **Launch incentives** for **long-distance trips**, especially during off-peak hours.
2. **Leverage real-time traffic data** for **efficient routing**, boosting trip turnover.
3. **Target high-revenue trips** (e.g., airport runs, cross-borough routes) in marketing and service plans.

LOCATION-BASED FLEET STRATEGY

1. **Prioritize Manhattan**, especially **East Harlem North & South**, using time-based zone deployment (e.g., AM in North, PM in South).
2. **Tiered coverage model**:
   - **Tier 1**: Continuous in high-demand zones (e.g., East Harlem, Central Park).
   - **Tier 2**: Rotational in moderate areas (e.g., Forest Hills, Fort Greene).
   - **Tier 3**: On-demand in low-volume borough (e.g., Bronx, Staten Island).
3. **Align fleet availability** with each zone's **hourly demand**, especially during **peaks**.

PASSENGER BEHAVIOUR AND SERVICE DESIGN

1. **Encourage ride-sharing** via apps, discounts, and eco-messaging in busy zones.
2. **Deploy large vehicles near airports, hotels, events**; support pre-booking for groups.
3. **Promote digital use** with incentives, while **keeping cash options** for inclusivity.

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